**Mock SaaS Architecture**
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2 **Programmatic MCP Edge Protection via Cloud Armor**

AI agents connect programmatically to MCP routes ( `/work-week/mcp/*` and `/service-immediately/mcp/*` ). These routes bypass IAP to support API clients, but are shielded by **Google Cloud Armor security policies** at the global Load Balancer edge. Cloud Armor applies rate-limiting, IP restrictions, and Web Application Firewall (WAF) rule sets to block brute-force and layer-7 DDoS attacks, protecting both the MCP endpoints and the database resources behind them before traffic reaches the containers.

3 **Private VPC Network & AlloyDB Isolation**

The Cloud Run container communicates with the database backend via a private Google Cloud VPC Network. Outbound queries route through a Serverless VPC Access connector to a private VPC Subnet, which connects via a VPC Peering Connection to the AlloyDB instance's primary IP ( `10.0.0.X` ). The database has no public IP address and is completely isolated from direct external internet access.

4 **HMAC Token Protection & Secret Manager**

Personal access tokens are verified statelessly using secure cryptographic HMAC signatures. The signing secret is managed in Google Cloud Secret Manager and accessed dynamically by the Cloud Run Service Account. This avoids storing credentials in plaintext and protects the token from intermediate interception or forgery, eliminating any local session database search overhead.

## Scalability & Performance

Designed to execute in high-traffic production environments, the platform implements a fully-managed serverless routing topology.

1 **Serverless Compute Autoscaling**

The core application runs on Google Cloud Run. Compute containers scale up dynamically from 0 to N instances based on CPU utilization and request concurrency, automatically scaling down to zero during inactive periods to minimize costs.

2 **Global Edge Routing Infrastructure**

The Global HTTPS Load Balancer routes traffic to serverless Network Endpoint Groups (NEGs). This setup connects clients directly to the nearest Google Frontend (GFE) edge server, reducing latency and defending against DDoS attacks.

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